

Package Insert

Sabah Oxygen MEDICAL AIR 100%v/v

1. Product Name

Sabah Oxygen MEDICAL AIR 100%v/v

2. Name and Strength of Active Substance(s)

Medical Air 100%v/v, Medicinal Gas, Compressed.

There are no excipients in this product.

3. Product Description

Medical Air is colourless and odorless. The product is stored in cylinders with a grey body and black and white shoulder. It is compressed gas at 150 bar at 27 Degree Celsius.

4. Pharmacodynamics

Pharmacotherapeutic Group- Medical Gas

ATC Code: V03AN05

The atmospheric air contains approximately 21% oxygen, 78% nitrogen and 1% trace gases. Similarly, medical air also contains approximately 21% oxygen, 78% nitrogen and 1% trace gases. Oxygen content in the medical air is the main reason for its usage as it is similar to the atmospheric air. Oxygen is essential for life and must be consistently delivered to all tissues to sustain cellular energy production. The ultimate destination for oxygen is the mitochondria within individual cells, where it is utilized in an enzymatic chain reaction to produce energy. Additionally, oxygen, in the form of reactive oxygen species, plays a crucial role in various immunological defense mechanisms.

Nitrogen gas is considered inert as it is in equilibrium within the total lung volume and has no pharmacological effects on the body.

5. Pharmacokinetics

Medical air contains approximately 21% oxygen, which is similar to the concentration of oxygen of atmospheric air. It is administered through the airways to the lungs, where gas exchange occurs in the pulmonary alveoli. This exchange happens down a partial pressure gradient, allowing oxygen from the inspired air/gas mixture to enter the capillary blood. Oxygen is then transported by the systemic circulation, primarily bound to hemoglobin, to the capillary beds in various tissues throughout the body. Only a small fraction of oxygen is dissolved freely in plasma. As blood passes through the tissues, oxygen is delivered to

individual cells based on partial pressure gradients. The oxygen absorbed by the body is almost entirely excreted as carbon dioxide, produced during intermediary metabolism. Nitrogen, on the other hand, is not absorbed and is exhaled without undergoing any conversion or metabolism.

6. Indications

Medical Air is used for:

- For normal respiration as replacement of atmospheric air.
- In anaesthetic equipment as a carrier gas during surgery.
- For nebulisation of inhaled drugs when oxygen supplementation is not required.

7. Recommended Dosage

Dosage

Please consult doctor for dosage. There is no distinction in the use of medical air between age groups.

Method of administration

For breathing purposes, medical air is administered by various means, commonly by self-contained or compressed air breathing apparatus.

In anaesthesia medical air is administered from a cylinder and valve assembly or pipeline through a face mask or endotracheal tube.

8. Route of Administration

Inhalation.

9. Contraindications

None known.

10. Warnings and Precautions

When mixed with other gases, Medical Air should only be administered to a patient if the oxygen content is at least 21%.

Medical Air should not be used near sources of ignition as it strongly supports combustion.

11. Interaction with other Medicaments

None known.

12. Pregnancy and Lactation

None known.

13. Side Effects

None known.

14. Symptoms and Treatment of Overdose

Not applicable.

15. Effects on Ability to Drive and Use Machines

None known.

16. Instruction of Use

Precautions and Preparation for Use

- Smoking and naked lights must not be allowed within the vicinity of cylinders or pipeline outlets.
- Cylinder valves and any associated equipment must never be lubricated and must be kept free from oil and grease.
- Medical gases must only be used for medicinal purposes.
- Remove the cylinder seal cap from valve before use.
- Cylinder valves should be opened momentarily prior to use to blow any foreign matter out of the outlet.
- Only the appropriate regulator should be used for the particular gas concerned.
- Never use excessive force when connecting equipment to cylinders.
- Cylinder valves must be opened slowly.
- Should leaks occur this will usually be evident by a hissing noise.

- There are no user serviceable parts associated with these valves, do not attempt to correct any problems with leakage from any part of the valve itself. Label any faulty containers, and return them to supplier for repair.
- Sealing or jointing compounds must never be used to cure a leak.

Use of Cylinders

- Cylinders should only be moved with the appropriate size and type of trolley.
- Cylinders should be handled with care and not knocked violently or allowed to fall.
- When in use cylinders should be firmly secured to a suitable cylinder support.
- When only a small amount of gas remains in a cylinder, the cylinder valve must be closed. It is important to leave a small residual pressure in each cylinder after use, in order to protect the inside of the cylinder from contamination.
- After use cylinder valves should be closed using moderate force only and the pressure in the regulator or tailpipe released.

Disposal

- Used cylinders should be returned to supplier immediately.

17. Storage Condition

Medical Air cylinders should be:

- Prohibiting smoking and naked flame in the storage area.
- Cylinders should be stored indoor, kept dry and clean.
- Kept away from any flammable materials.
- Store in a well ventilated place.
- Store in upright position and secure it.
- Full and used cylinders should be stored separately. Full cylinders should be used in strict rotation.
- Medical cylinders containing different gases should be segregated and identified within the store.
- Cylinders must not be repainted, have any markings obscured or labels removed.
- Store at temperature below 50 °C.

18. Shelf Life

3 years

19. Contents of Container

Medical Air is supplied as compressed gas stored in steel cylinders equipped with BullNose valves.

Cylinder Size (L water capacity)	Content (m ³ °C at 27,1bar)	Cylinder Material	Valve	Pressure (bar)
41	6.4	Steel	BullNose	150
47	7.3	Steel	BullNose	150

20. Name and Address of Manufacturer/product register holder

SABAH OXGEN SDN. BHD.

Lot No. 30, Jalan 2A KKIP Timur, Industrial Zone 12 (IZ12), Kota Kinabalu Industrial Park (KKIP), 88460 KOTA KINABALU SABAH, MALAYSIA

21. Product Registration Number

MAL24126038X

22. Date of Revision of the Text

3/6/2024